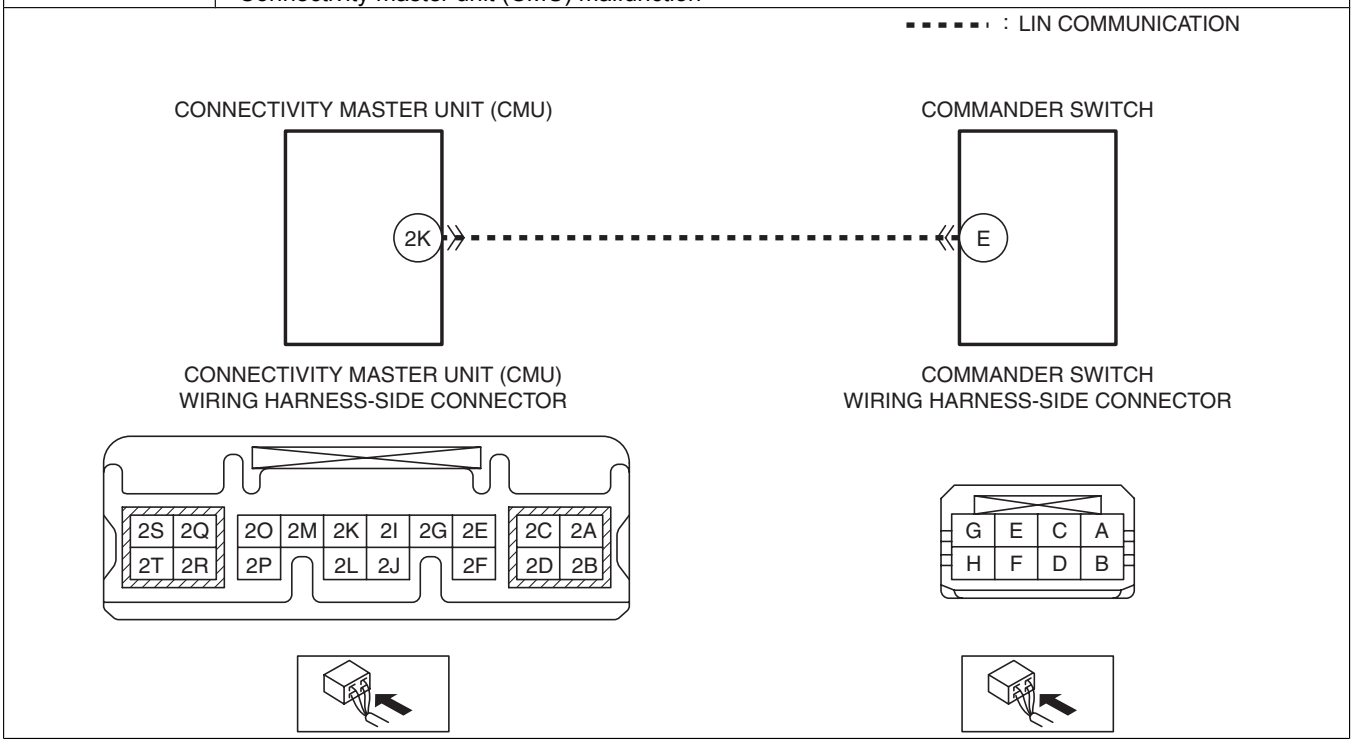
DTC B1087:87 [CONNECTIVITY MASTER UNIT]

id0902n7103100

Description	Communication error with commander switch
Detection condition	• The connectivity master unit (CMU) cannot receive a LIN signal from the commander switch for **5 s or more** with the ignition switched ACC or ON (engine off or on).
Fail-safe	Not applicable
Possible cause	• Commander switch connector or terminal malfunction • Connectivity master unit (CMU) connector or terminal malfunction • Short to ground in wiring harness between connectivity master unit (CMU) terminal 2K and commander switch terminal E • Short to power supply in wiring harness between connectivity master unit (CMU) terminal 2K and commander switch terminal E • Open circuit in wiring harness between connectivity master unit (CMU) terminal 2K and commander switch terminal E • Commander switch malfunction • Connectivity master unit (CMU) malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT COMMANDER SWITCH CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the commander switch connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.

Step	Inspection		Action
2	**INSPECT CONNECTIVITY MASTER UNIT (CMU) CONNECTOR CONDITION** • Disconnect the connectivity master unit (CMU) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.
3	**INSPECT COMMANDER SWITCH CIRCUIT FOR SHORT TO GROUND** • Verify that the commander switch and connectivity master unit (CMU) connectors are disconnected. • Inspect for continuity between commander switch terminal E (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify if there is a common connector between connectivity master unit (CMU) terminal 2K and commander switch terminal E. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground. Go to Step 7.
		No	Go to the next step.
4	**INSPECT COMMANDER SWITCH CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the commander switch and connectivity master unit (CMU) connectors are disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at commander switch terminal E (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify if there is a common connector between connectivity master unit (CMU) terminal 2K and commander switch terminal E. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to power supply to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to power supply. Go to Step 7.
5	**INSPECT COMMANDER SWITCH CIRCUIT FOR OPEN CIRCUIT** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Verify that the commander switch and connectivity master unit (CMU) connectors are disconnected. • Inspect the wiring harness for continuity between connectivity master unit (CMU) terminal 2K (wiring harness-side) and commander switch terminal E (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify if there is a common connector between connectivity master unit (CMU) terminal 2K and commander switch terminal E. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 7.

Step	Inspection		Action
6	**VERIFY IF MALFUNCTIONING LOCATION IS COMMANDER SWITCH DEPENDING ON REPEATABILITY** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the connectivity master unit (CMU) using the M-MDS. (See CLEARING DTC [CONNECTIVITY MASTER UNIT].) • Switch the ignition ON (engine off or on) and wait for **12 s or more**. • Retrieve the connectivity master unit (CMU) DTCs using the M-MDS. (See DTC INSPECTION [CONNECTIVITY MASTER UNIT].) • Is the same DTC displayed?	Yes	Replace the commander switch, then go to the next step. (See COMMANDER SWITCH REMOVAL/ INSTALLATION.)
		No	Go to Step 8.
7	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the connectivity master unit (CMU) using the M-MDS. (See CLEARING DTC [CONNECTIVITY MASTER UNIT].) • Switch the ignition ON (engine off or on) and wait for **12 s or more**. • Retrieve the connectivity master unit (CMU) DTCs using the M-MDS. (See DTC INSPECTION [CONNECTIVITY MASTER UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the connectivity master unit (CMU). (See CONNECTIVITY MASTER UNIT (CMU) REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
8	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [CONNECTIVITY MASTER UNIT].)
		No	DTC troubleshooting completed.